

A Wearable Sensor Based Multi-Criteria-Decision-System for Real-Time Seizure Detection

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Abstract—This paper presents a wireless, low power and low cost two part wearable for real-time epileptic seizure detection. Using parameters of Electro-cardiograph (ECG), Electro-dermal Activity (EDA), body motion and breathing rate (BR), a novel multi-criteria-decision-system (MCDS) is proposed that reduces false alarms and true negatives. The combination of a chest and hand worn wearable continuously senses these parameters transmitting the data to a smart phone application via BLE 4.0 where long-short-term-memory (LSTM) based anomaly detection algorithms and logistic classifiers decide on the occurrence of the seizure in real time. A 96% precision and 90% recall is achieved through testing on synthetic data.

I. INTRODUCTION

An estimated \$25000 is spent by US households on healthcare yearly being a huge financial burden both on individuals and governments. A major portion of this cost is due to hospitalization and regular follow-up visits. To mitigate this issue, wearable sensors provide a great opportunity to measure human physiology in a remote, continuous, real-time and non-invasive manner. Such sensors, when used in ambulatory settings, have the potential to reduce the above mentioned costs many folds. These solutions are especially relevant for chronic diseases such as Alzheimer, Parkinson, Diabetes and Epilepsy.

Epilepsy is the third most common neurological disorder with 65 million patients and more than 200,000 new patients being diagnosed every year [1]. It is caused by abrupt and abnormal brain nerve cell activity, causing involuntary body movements termed as seizures. A seizure, depending on the source of origin in the brain can be categorized as a generalized or partial seizure (also called local or focal respectively). A seizure can cause the patient to lose consciousness, followed by generalized body stiffening and violent jerking after which the patient enters the post ictal phase of a deep sleep [2]. Seizures can cause head injuries, fractures, cuts and bruises, loss of blood if unattended and even burn wounds. Up to 50,000 deaths occur in US every year due to Epilepsy due to drowning, accidental deaths, choking on vomit and SUDEP (Sudden Unexpected Death due to Epilepsy).

The unpredictability of seizure occurrence is a major issue posing threat, not only to the patient but also to the family. Thus, an Epileptic patient needs to be under the constant supervision of a nurse or a caretaker, making it both costly and a time intensive activity. These circumstances indicate a dire need for reliable and continuous monitoring of epileptic patients.

The gold standard for the monitoring and diagnosis of seizures is the electroencephalogram (EEG) along with camera based video feed. But this technique poses its own challenges of hospital setting, with costly equipment and technicians for

analysis. Moreover, prolonged recordings with multiple electrodes placed on the patient can be uncomfortable. Lastly, the post analysis hinders it from being a real-time method.

This has motivated researchers to look towards ambulatory solutions among which the most common method is to sense the random wrist movements using motion sensors. One such example is demonstrated in [3], where 7 out of 8 seizures are correctly detected but at the same time, 204 non seizure activities are also labeled as seizures. Another study in [4], claims only 48% seizures being identified with 3D accelerometers. Thus, accelerometer based seizure detection can generate true negatives and is prone to generate false alarms in activities like brushing or repetitive hand movement.

Similarly, some other studies have used patient ECG, or combination of EEG & ECG for seizure detection [5]. Another rather new approach being adopted is use of the electro-dermal activity (EDA) or the Autonomic Nervous System (ANS) response for detection purposes [6]. However, in using any of these techniques for seizure detection, the issue is the appearance of the same anomalies in other daily activities thus producing false alarms. For example, tachycardia is eminent during a seizure but also happens during running or in any other physical activity. Similarly, EDA response is prominent during seizures but also in sleep or any tense, happy or fear situation.

In order to address the above issues, this paper utilizes the fusion of four physiological parameters to reduce false alarms. A two part wearable measuring EDA, ECG, BR and motion sensing is described with a companion android application with real-time anomaly detection and classification. Subsequently, tests of healthy subjects in everyday situations and results of a seizure situation using synthetic data are also presented.

II. DESIGN METHODOLOGY/IMPLEMENTATION

The overall block-diagram of the proposed system is shown in Fig. 1. It consists of two wearable parts consisting of a wrist band and a chest band. The former consists of a 3D accelerometer to sense the repetitive involuntary wrist motion.

Furthermore, the wrist band also contains Ag/AgCl electrodes to pick up the EDA signal. On the other hand, the chest wearable contains three Ag/AgCl electrodes carefully positioned for the acquisition of the ECG signal. Moreover, a one inch by one inch Force Sensitive Resistor (FSR) is deployed to monitor the respiration rate. Both wearables are powered up by separate 3.7V Li-Ion batteries. The wrist module has a 500mAh battery and with a current consumption of 7mA is capable of running for continuous 3 days. Similarly, the chest module has a 600mAh battery with a battery life of 3 days. The modules are rechargeable via a micro-USB cable.

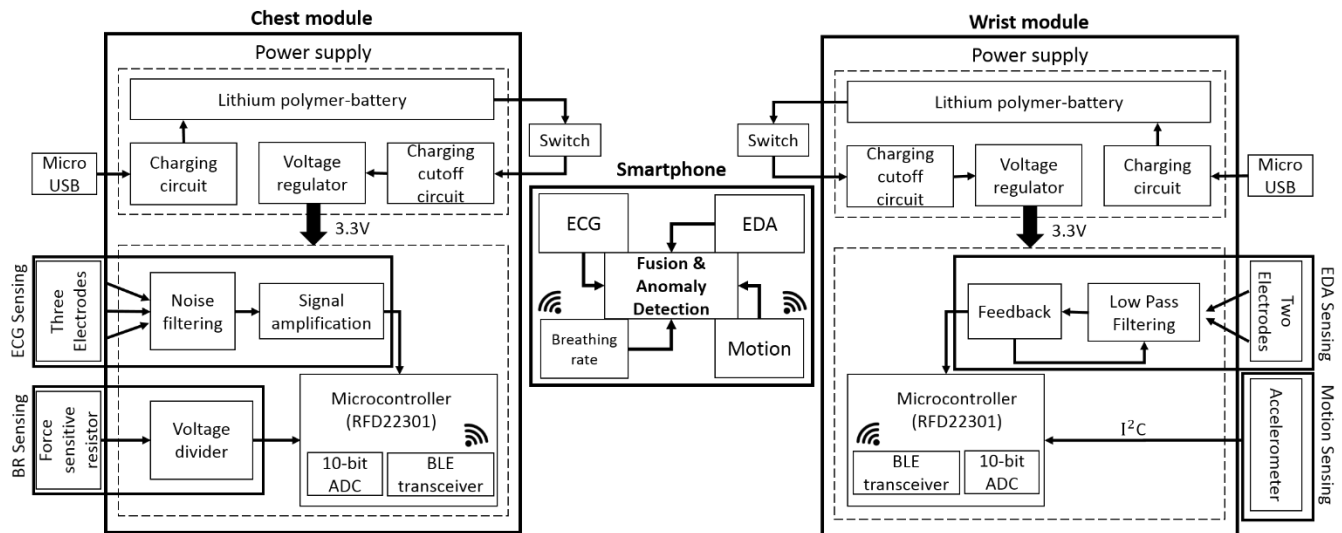


Figure 1: Overall system block diagram

The micro-controller used for the data acquisition, analysis and communication with the smartphone is RFD22301 chip. It has built in Bluetooth Low Energy (BLE) capability. It samples analog signals at a frequency of 100Hz and digitizes the signal using a 10-bit Analogue to Digital Converter (ADC). Each module has its own microcontroller chip with each of them communicating simultaneously to a smart phone. In case of anomaly in the data the patient's smartphone marks it as a seizure and generates an alarm to notify people nearby and also generate a notification at the caretaker app.

The wearables are 3D printed in light weight, miniaturized and ergonomic casings. The wrist part had two compartments of 38x38x18 and 28x17x12 mm³ each, making it like a smart watch. The chest wearable with its adjustable strap and curved design is easy to wear and adaptable to different chest sizes.

A. ECG

Different anomalies occur in the ECG at the onset of seizure. For instance, Ictal-onset tachycardia [7], sinus tachycardia [8], sinus arrhythmia, sinus pause, premature atrial depolarization, AV-block, bundle branch block, T wave inversion and ST elevation or depression [9].

To acquire the ECG, the designed circuit consists of five stages: Instrumentation amplifier, high pass filter, notch filter, low pass filter and DC level shifting. Instrumentation amplifier is used to amplify low input signal because of its high input impedance and high CMRR. High pass filter with the cutoff frequency of 0.05 Hz removes any DC component in the signal, notch filter of 50 Hz gets rid of the power line noise and the low pass filter having cutoff frequency of 150 Hz filters the white noise. Finally, the DC level is shifted to

1.65V (half of the V_{CC}) in order to get the full swing of the signal.

B. EDA

EDA also known as skin conductance is an objective index of human emotional state signified with sweat at specific body areas. Multiple studies have been carried out to study the relation between EDA and seizures, showing significant deviations from the baseline EDA at seizure onset [10] [11].

The proposed EDA circuit consists of two stages, the first one is a 2Hz active low pass filter; removing any noise from the signal and the second stage is a feedback voltage follower for automatic biasing, to cater for different skin conductance levels. As the eccrine sweat glands are responsible for EDA response the electrodes are placed at the distal region of the wrist. This is also supported by the study in [12], suggesting that EDA is prominent on fingers, feet, shoulder, neck & wrist.

C. Motion Sensing

The epileptic patient experiences severe abnormal motion while occurrence of a seizure. These motions can occur at different body locations depending on the seizure type and origin in brain. The motion sensing is implemented using 3-axis accelerometer ADXL-345. The motion has its maximum manifestation on the limbs hence the accelerometer is applied in the wrist module. The jerks can be as much as of 10g acceleration, which is well within the 16g range of the sensor. A sampling rate of 100 Hz is used which is also greater than twice the 25Hz maximum frequency of limbs motion.

D. Respiration

At the onset of seizure the patient encounters remarkable variation in BR, thus it can also serve as a diagnostic parameter. In a research conducted among 21 patients, 19 experienced tachypnea and 1 experienced bradypnea [13]. The form-factor of the chest module provides an edge in implementing the BR sensing through a very simple approach. A force sensor FSR-406 of Interlink Electronics is placed at the bottom of the chest module, pressed against the chest while the patient inhales. Thus, providing breathing pattern in voltage division circuitry.



Figure 2: Wrist and Chest Wearables

E. Software

Seizures are detected by anomaly detection in ECG, EDA, BR and Motion. The anomaly in ECG and BR were detected by LSTM Networks that were trained to predict the next sample in the sequence. The finalized architecture contains a single hidden layer with 16 neurons. Sliding window of input is taken with 30 samples which are divided into 10 timestamps with three samples at each timestamp. The 31st sample in the sequence served as the target. The network works as a regression unit producing a continuous value at the output, predicting the value of next sample in sequence. The network was trained using Mean-squared error loss function.

Since movement that the patient is engaged in during an epileptic seizure was already known from literature, a network was trained in a fully-supervised manner. The network architecture for movement classification was very similar to the one used for ECG and BR except that the final output wasn't a continuous function now. Therefore, softmax layer was added at the top of output with two units since there were two classes in the data i.e. Normal Movement and Seizure Movement. Input was fed in as a sequence of 30 samples at 10 timestamps with three samples at each time step. Label was encoded using one-hot encoding with 2 classes. The network was trained using Cross-Entropy loss function.

As EDA is a slowly varying signal, a simple statistical anomaly detection system was used which generated an anomaly score if the sample value was three standard deviations away from mean.

All these anomaly scores were then combined using a logistic classifier. Linear combination of anomaly scores followed by threshold operation on final value produced similar results. The classification algorithm block diagram is shown in Fig. 3. On top of the algorithms, separate sampler, caretaker and Web apps were built to allow the caretaker and doctor to view the relevant physiological data and keep track of seizures. Moreover, in case of seizure occurrence, patient's GPS coordinates are sent in an alert message to the caretaker using GSM communication.

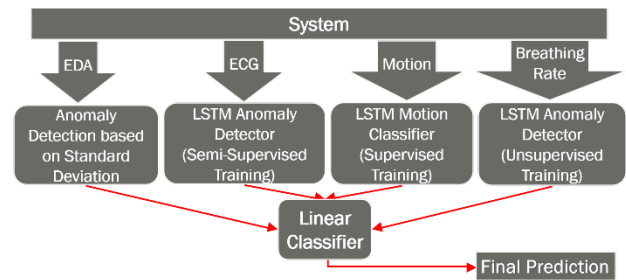


Figure 3: Anomaly detection algorithm overview

III. RESULTS

All parameters of 4 healthy individuals were monitored, with written informed consent of each participant. The time domain waveforms of a single individual's parameters under different scenarios are shown in Fig. 4. In case of acceleration, sitting shows minimal movement, with a rapid to & fro motion in walking and a slow to & fro motion in smoking. For EDA, an obstacle avoidance game, with increasing difficulty level, was played by the subject. Slowly, increasing EDA is visible with surges in different situations during the game. The signal is stable during relax mode and peaks appear during sleep signifying the ANS activity. For ECG, a normal heart rate of ~72 beats/min is visible in relax mode with the heart rate increasing during physical exertion. Similarly, normal BR is observed in relax mode with a high BR while running and low when sleeping.

As a first step in this research, synthetic data was used to simulate the seizure condition. Accelerometer's data signifying patient's movement is shown in Fig. 5 (a). The data is fed to the classifier and the framework then generates probabilities for indication of sequence belonging to that particular class. Similarly, Fig. 5 (b) shows the results for EDA signal with red lines indicating anomalies detected by the system.

Anomaly detection in BR and ECG is shown in Fig. 6 (a)-(b). The input sequence is indicated in blue while the predictions generated by the system are shown in green. The discrepancy between original sample and the predicted sample is indicated in red. The Squared error will be high in case of anomaly and low in case of non-anomalous points.

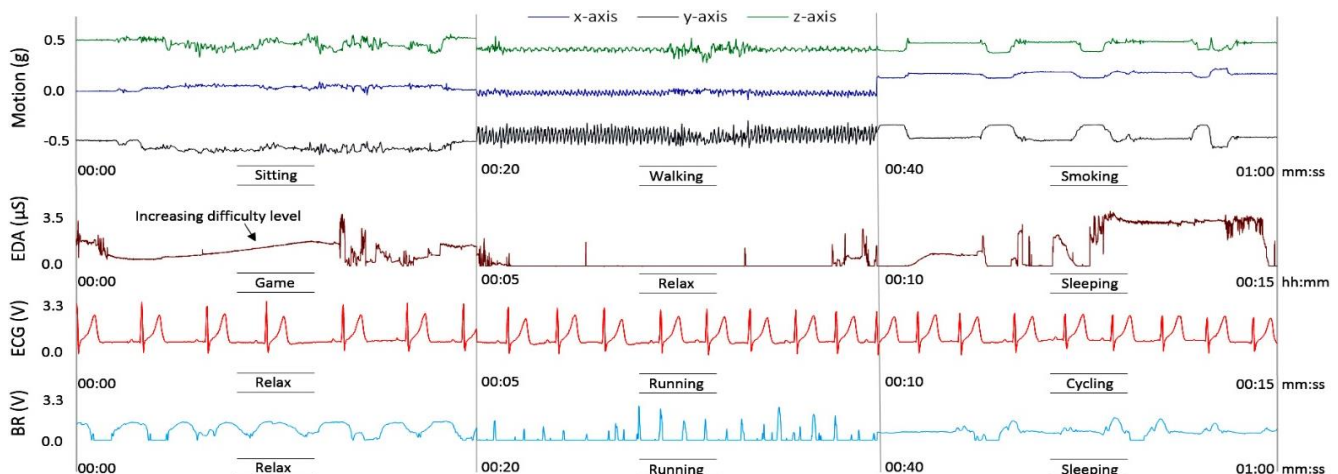


Figure 4: Motion, EDA, ECG & BR time domain signals under different situations

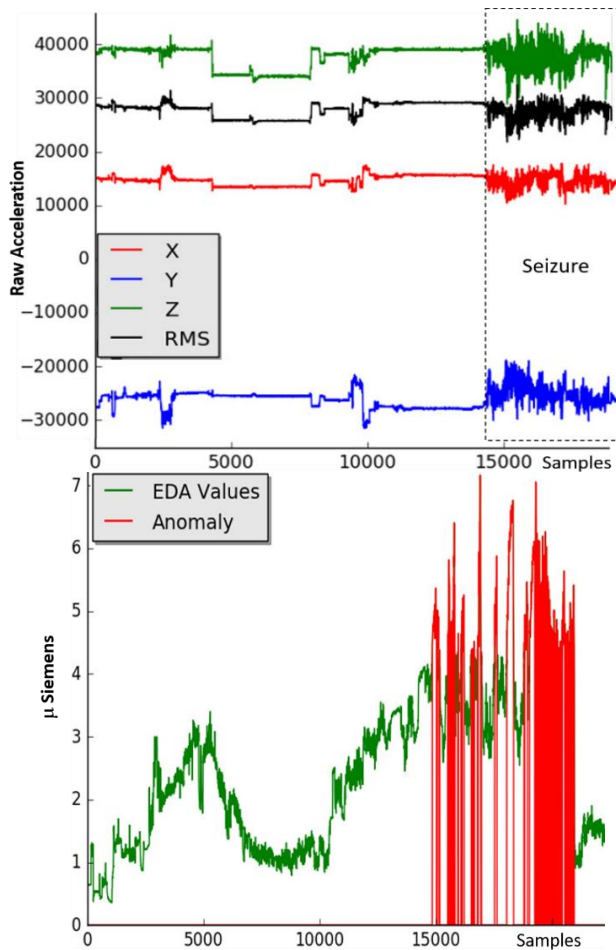


Figure 5: a) Motion data and b) EDA data anomalies

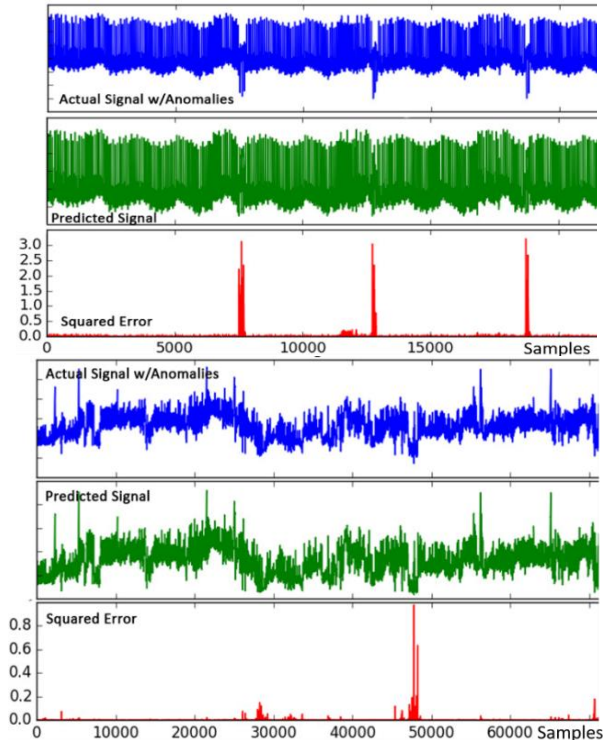


Figure 6: a) ECG data and b) BR data anomalies

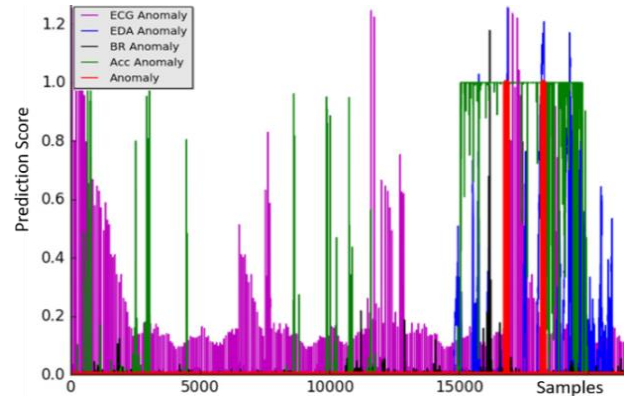


Figure 7: Final seizure prediction graph

These raw anomaly scores were then fed to a linear classifier. The scaled anomaly values are shown in Fig. 7. Different parameters had high anomaly scores at different times but the final classifier predicted seizure only when all the factors detected anomalies simultaneously, creating a more robust system with a 96% precision and 90% recall.

IV. CONCLUSION

A Multi-Criteria-Decision-System, composed of two hardware wearables & anomaly detection software framework, for accurate epileptic seizure detection is presented. In contrast to earlier works, which only use 1-2 parameters, this work utilizes four physiological parameters reducing the false alarms and true negatives. Furthermore, in comparison to EEG, the 3 day battery life makes it a viable solution for ambulatory monitoring of epileptic patients.

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